UNIT 10 · TOPIC 10

Tractor–Implement Interface Technology (ISOBUS)

Syllabus topic: Tractor-implement interface technology

ISOBUS (ISO 11783)

- A CAN-bus network that standardises communication between tractor and implement.
- A single **Universal / Virtual Terminal** in the cab controls any maker's ISOBUS implement.

The Task Controller

- Performs automatic **SECTION CONTROL** — shuts off boom sections/rows over already-treated or out-of-field areas, eliminating overlap and input waste.
- Performs **VARIABLE-RATE** control from the prescription map.
- Logs as-applied data (Topic 8).

GPS auto-guidance/auto-steer (RTK — Topic 3b) keeps passes precise. This integration is what makes site-specific application practical in the field.

Practice MCQs

Q60. The standard tractor–implement communication network is:

(a) ISOBUS (ISO 11783) (b) ISO 9001
(c) HDMI (d) RS-232

Q61. Automatic shut-off of boom sections over already-treated areas is:

(a) Section control (b) Yield mapping
(c) Kriging (d) Calibration

Q62. Section control mainly reduces:

(a) Fuel octane (b) Input overlap and waste
(c) Engine noise (d) Tyre wear

Q64. The ISOBUS component that performs variable-rate and logging is the:

(a) Universal Terminal only (b) Task Controller
(c) Alternator (d) Governor

Answers

Q60 — (a) ISOBUS (ISO 11783) — *ISOBUS (ISO 11783).*

Q61 — (a) Section control — *Task Controller section control cuts overlap.*

Q62 — (b) Input overlap and waste

Q64 — (b) Task Controller — *Task Controller does variable rate and logging.*

Sources: Srinivasan — Handbook of Precision Agriculture; ICAR/NIIT PA modules; Lillesand & Kiefer — Remote Sensing and Image Interpretation; ISO 11783 (ISOBUS); remote-sensing and NDVI references (as given in the source notes, July 2026); NPTEL precision-agriculture lectures.